

Concept Mapping – OSI Model

Data Units, Layers, and Addressing for Reliable Communication

Question 1: Missing Layers in the Concept Map

Definitions

Session Layer (L5): Establishes, manages, and terminates the communication session (dialogue) between two applications. It handles dialogue control (deciding whose turn it is to send data) and synchronization, inserting checkpoints in long transfers so that only data after the last checkpoint needs to be resent if a failure occurs.

Data Link Layer (L2): Provides node-to-node delivery of data within the same local network. It organizes bits into frames, attaches MAC addresses for local identification, performs error detection (e.g., using CRC), and manages access to the shared physical medium.

Problem Statement

The given concept map shows the sequence: Application → Presentation → ___ → Transport → Network → ___ → Physical. Two layers are missing from this chain. Identify these layers and explain why each one's function is essential for reliable communication.

Analysis

Blank 1 = Session Layer — sits between Presentation and Transport, and coordinates and synchronizes the dialogue between communicating applications so that data exchange stays organized and recoverable.

Blank 2 = Data Link Layer — sits between Network and Physical, and converts packets into frames, adds MAC addressing, and checks for transmission errors before data is placed on the physical medium.

Without the Session Layer, there would be no way to track which side should be transmitting or to recover cleanly from an interrupted session, since no checkpoints or dialogue control would exist.

Without the Data Link Layer, packets could not be safely placed on a shared medium, since there would be no local addressing, framing, or error detection to catch corrupted data before it reaches the physical wire or signal.

Together, these two layers close the gap between application-level communication and raw signal transmission, which is why the full seven-layer chain is needed for dependable, end-to-end communication.

Concept Map

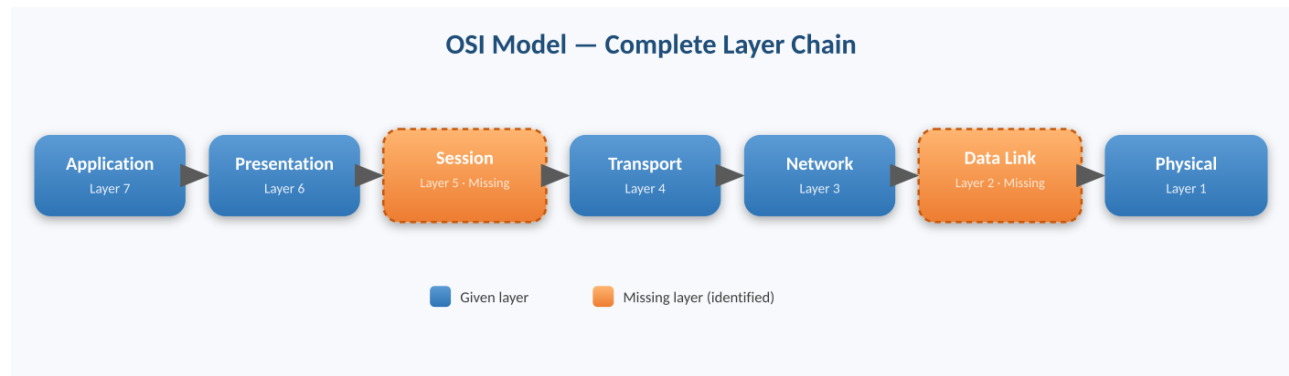


Figure 1: Completed OSI concept map with the Session and Data Link layers restored.

Question 2: Data Units Across OSI Layers

Definitions

PDU (Protocol Data Unit): The general term for the specific form data takes at each OSI layer as it is processed. Every layer encapsulates the data received from the layer above by adding its own header (and sometimes a trailer), producing a new named data unit.

Segment: The Transport Layer's PDU, formed by adding a header with information such as port numbers and sequence numbers (as in TCP) to the raw data.

Packet: The Network Layer's PDU, formed by adding an IP header containing source and destination IP addresses to the segment.

Frame: The Data Link Layer's PDU, formed by adding a header and trailer containing MAC addresses and error-checking information to the packet.

Bits: The Physical Layer's PDU — the frame converted into a raw stream of 0s and 1s for transmission as electrical, optical, or radio signals.

Problem Statement

Construct a concept map linking the data units Segment, Packet, Frame, and Bits to their respective OSI layers, and analyze how this transformation of data units supports transmission.

Analysis

As data descends the OSI stack, each layer encapsulates it by attaching its own control information, which is why the data unit's name changes at every layer: Data → Segment → Packet → Frame → Bits.

This progressive transformation allows each layer to perform its job independently — the Transport layer handles reliability and ordering, the Network layer handles addressing and routing, the Data Link layer handles local delivery and error checking, and the Physical layer handles the actual signal transmission.

At the receiving end, the reverse process (de-encapsulation) strips away each header in turn, allowing the original data to be reconstructed accurately.

This layered transformation means problems at one layer (such as a corrupted frame)

can be detected and handled without every other layer needing to understand the full complexity of the transmission, supporting modular and reliable data transfer.

Concept Map

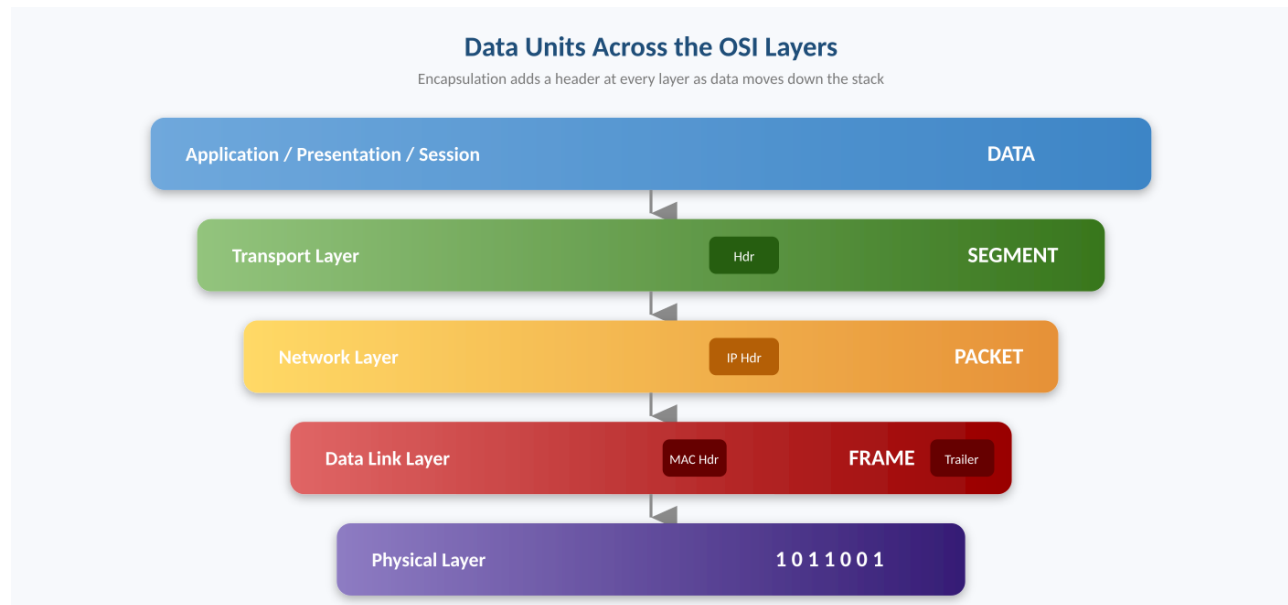


Figure 2: Encapsulation of data into Segment, Packet, Frame, and Bits across the OSI layers.

Question 3: MAC Address and IP Address Delivery Roles

Definitions

MAC Address (Media Access Control Address): A physical address burned into a device's network interface card, used by the Data Link Layer to identify devices uniquely within the same local network segment.

IP Address (Internet Protocol Address): A logical address assigned by the Network Layer, used to identify a device across interconnected networks and enable routing of data toward a distant destination network.

Problem Statement

Given the concept map Data Link → MAC Address → Local Delivery and Network → IP Address → Global Delivery, explain how these relationships ensure complete data delivery across networks.

Analysis

The Data Link Layer uses the MAC address to achieve local delivery — moving a frame from one device to the next directly connected device, such as across a single LAN segment or between a host and its router.

The Network Layer uses the IP address to achieve global delivery — determining the overall path a packet must take from its source network to its final destination network, potentially crossing many intermediate routers.

Global delivery depends on repeated local deliveries: at every hop along the route, the

IP address tells the network where to send the packet next, while the MAC address at that hop tells the local segment who to hand it to right now.

The IP address stays constant from source to destination, while the MAC address changes at every hop as the packet is re-framed for the next local segment — the two addressing systems working together to guarantee complete, accurate, end-to-end delivery.

Concept Map

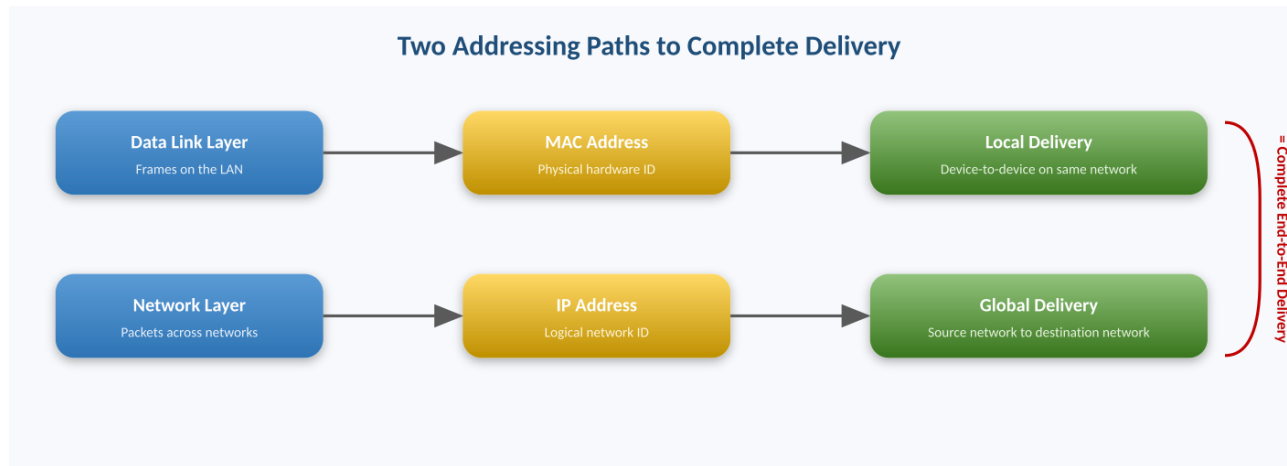


Figure 3: MAC-based local delivery and IP-based global delivery combining for end-to-end delivery.